SAM-Prior-Guided Spatial–Spectral–Temporal Mamba Framework for Affine Misregistration Hyperspectral Change Detection

Tongzhen Zhang, *Student Member, IEEE*, Wenqian Dong, *Senior Member, IEEE*, Yunsong Li, *Member, IEEE*, and Jiahui Qu, *Senior Member, IEEE*

***Abstract*—The hyperspectral change detection (HSCD) aims to identify land surface changes by analyzing spatial and spectral differences between multitemporal hyperspectral images (HSIs). However, most existing methods are designed under the assumption of strict pixel-level correspondence, making them intrinsically sensitive to affine misregistration that frequently occurs in real-world scenarios due to sensor drift, platform instability, or accumulated georeferencing errors. To address this issue, we propose an HSCD framework that integrates image registration and change detection through object-level structural priors. Specifically, the segment anything model (SAM) is employed to extract semantically consistent region masks from pseudo-color projections of HSIs. These masks guide both object-level alignment and sparse label propagation, enabling structure-aware supervision without requiring dense annotations. The expanded labels, encoding object-level priors, are then used to supervise a spatio-spectral-temporal change detection network built upon convolutional and Mamba modules. This architecture effectively models spatial, spectral, and temporal dependencies while maintaining robustness against affine misregistration. Extensive experiments on three HSCD benchmarks demonstrate that our method consistently outperforms existing approaches under both well-aligned and affinely misregistered conditions. The code is available at: https://github.com/Jiahuiqu/SAMPG-SST-MCDNet**

***Index Terms*—Change detection (CD), hyperspectral image (HSI), Mamba, segment anything model (SAM), unregistered scenarios.**

I. Introduction

CHANGE detection (CD) is a fundamental task that aims to identify land surface changes by comparing multitemporal remote sensing (RS) images [1], [2]. It supports a wide range of applications, including land cover monitoring, urban development analysis [3], and disaster assessment [4]. Hyperspectral image (HSI) provides rich spectral information across hundreds of contiguous bands, enabling fine-grained material discrimination [5], [6]. Compared with RGB or multispectral data [7], HSI is more sensitive to subtle spectral differences, making it particularly valuable for detecting complex or low-contrast changes. This advantage has motivated growing interest in hyperspectral change detection (HSCD) [8], [9].

A wide range of methods have been proposed for HSCD, spanning from traditional models to modern deep learning-based approaches. Traditional methods, such as change vector analysis (CVA) [10], principal component analysis (PCA) [11], and conditional random fields [12], often rely on handcrafted spectral difference measures. These approaches are simple and interpretable. However, they are sensitive to noise and lack the ability to capture robust and deep semantic features, limiting their effectiveness in complex change detection tasks. In recent years, deep learning-based methods have achieved notable success by learning more robust change representations [13], [14], [15], [16]. Siamese convolutional networks [17], [18], 3-D CNNs [19], RNN [20], and Transformer-based architectures have been introduced to capture spectral–spatial dependencies between bitemporal HSI pairs [21], [22]. Compared with handcrafted approaches, deep learning methods significantly improve performance by exploiting spectral–spatial context. While Transformer is capable of capturing long-range dependencies via self-attention, its quadratic computational complexity restricts efficiency on high-dimensional hyperspectral data. To address this, the Mamba architecture [23], built upon selective state-space models (SSMs) [24], [25], offers a robust solution with linear complexity and hardware-aware optimization [26]. Mamba is specifically suitable for HSCD as it efficiently encodes global spectral–spatial context and subtle material-level changes without the computational overhead of self-attention [27]. For instance, Chen et al. [28] proposed GDAMamba to enhance the global modeling. Zhan et al. [29] hyperspectral multiclass change detection under limited small-category samples. Ding et al. [30] combined wavelets with Mamba (WDP-Mamba) to preserve spatial structure. Wu et al. [23] introduced SSMIF to optimize feature alignment, and Yang et al. [31] developed the graph-based ARCD framework for limited-label conditions.

Despite their advances, most existing HSCD methods assume strict pixelwise alignment between input images. However, in real-world scenarios, geometric misalignment

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The authors are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: tzzhang_1@stu.xidian.edu.cn; wqdong@xidian.edu.cn; ysli@mail.xidian.edu.cn; jhqu@xidian.edu.cn).

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is common due to sensor drift, terrain variation, or platform instability. Such spatial inconsistencies lead to incorrect pixel correspondences and false spectral differences, significantly degrading the change detection performance. In the conventional two-stage workflow, registration is performed first, and change detection is subsequently applied, which inevitably transfers and accumulates registration errors into the detection phase, making the final accuracy overly dependent on registration quality. This motivates the development of alignment strategies explicitly tailored to the change detection task rather than treating registration as an isolated preprocessing step. By integrating spatial alignment directly into the change analysis pipeline, one can reduce the ambiguity between genuine land-cover variations and artifacts introduced by misregistration. In this context, several recent studies have explored joint registration and change detection frameworks to address the problem of unregistered change detection. Jing et al. [32] proposed the ChangeRD framework, which provides an important solution for unregistered change detection by integrating image registration and change detection into a unified multitask network. ChangeRD employs an adaptive perspective transformation (APT) module and an attention-guided central difference convolution (AgCDC) module to jointly learn alignment and change regions. While offering an efficient end-to-end solution, it primarily relies on pixel-level geometric transformations without explicit semantic structural modeling or deep supervisory coupling, limiting its robustness in complex scenes and its ability to exploit high-dimensional hyperspectral correlations. In contrast, our method establishes a tightly coupled "registration–supervision–detection" paradigm by leveraging SAM-based structural priors. By combining region-level alignment with spatial–spectral–temporal modeling, we ensure structure-aware supervision, achieving a transition from geometry-driven integration to semantically coherent task collaboration.

To operationalize this structure-aware paradigm, we introduce object-level priors into HSCD by leveraging the segment anything model (SAM) [33]. SAM provides stable region masks that delineate coherent structures across time, enabling more accurate and semantically meaningful alignment. These masks allow us to estimate fine-grained affine transformations that precisely register bitemporal HSIs. Importantly, the obtained object-consistent alignment not only reduces false alarms caused by misregistration but also facilitates label expansion within corresponding regions, providing more reliable supervision for training the change detection network.

While the incorporation of object-level priors improves spatial consistency and annotation efficiency, HSCD still faces intrinsic challenges due to the high spectral dimensionality and temporal dynamics. HSCD requires jointly modeling spatial structures, spectral correlations, and temporal transitions between bitemporal images. To this end, we propose a dual-branch spatio-spectral contextual encoder (SSCE) combined with a center-aware temporal difference decoder (CTFD), forming a unified framework tailored for HSCD. The SSCE module is designed to extract global–local spatial and spectral dependencies through directional Mamba modules and convolutional layers, enabling complementary modeling of long-range context and local semantics. The CTFD module integrates bitemporal features through interleaved temporal sequences, guided by center-aware difference attention that emphasizes difference features consistent with the central target, while suppressing the irrelevant background variations. Trained with structure-enhanced supervision from the SAM-guided label expansion, the proposed network fully exploits the spatial, spectral, temporal, and object-level information embedded in hyperspectral imagery, yielding robust and accurate change detection results under misalignment and annotation scarcity. In summary, the main contributions are as follows.

1) We propose an SAM-Mamba guided framework for HSCD under spatial misalignment, designed to leverage object-level priors for robust detection despite imperfect registration.
2) We propose an object-level registration method based on the segmentation masks generated by SAM. By leveraging these masks to guide region matching and affine transformation estimation, our method achieves structure-consistent alignment for bitemporal HSIs.
3) We propose the spatio-spectral-temporal Mamba change detection network (SST-MCDNet), which integrates convolutional layers and spectral–spatial Mamba blocks within a dual-branch encoder and introduces a CTFD. This design enables joint modeling of spatial, spectral, and temporal dependencies, enabling the precise pixel-level change detection.

The remainder of this article is organized as follows. Section II reviews related work in HSCD and foundation model adaptation. Section III introduces the overall architecture and components of the proposed method. Section IV presents the experimental results and relevant analysis. Section V concludes this article and outlines the future research directions.

II. Related Works

A. Image Registration

RS image registration aims to align images captured at different times, from different sensors, or from different viewpoints. Traditional methods for image registration generally include feature-based, intensity-based, and region-based approaches. Feature-based methods, such as scale-invariant feature transform (SIFT) [34] and speeded-up robust features (SURFs) [35], rely on detecting keypoints and matching them between images. While these methods perform well for global translational alignment, they are less effective when faced with large-scale misalignments or complex transformations. Intensity-based methods, such as mutual information (MI) [36], estimate geometric transformations by comparing pixel intensities. These methods can handle sensor differences effectively but may struggle with large displacements or regions lacking distinctive features. Region-based methods, like normalized cross correlation (NCC) [37], align images by evaluating the similarity of local pixel intensities. While these methods are computationally efficient, they are sensitive to noise and scene changes, limiting their application to high-resolution RS images.

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In recent years, deep learning-based methods have made significant advancements in RS image registration by learning powerful feature representations directly from the data [38]. Convolutional neural network (CNN)-based methods, such as DeepReg [39] and Voxelmorph [40], learn hierarchical features from image pairs and use regression networks to predict spatial transformations. These methods perform well in capturing complex patterns and large displacements, but they typically require large amounts of labeled data for training. Transformer-based methods, like AttentionReg [41], leverage self-attention mechanisms to model long-range spatial dependencies, offering significant improvements in handling larger transformations. However, these methods are computationally expensive, which limits their scalability for large-scale RS image applications.

Despite the significant progress in deep learning-based registration, most existing methods fail to incorporate change detection tasks into the registration process. The lack of synergy between the two tasks limits the ability to handle misalignment in a more context-aware manner, especially when considering the semantic-level features that are crucial for reliable change detection in RS images.

B. Segment Anything Model

The SAM is a large-scale vision foundation model designed for image segmentation. It adopts a three-component architecture consisting of an image encoder based on a Vision Transformer (ViT) [42], a lightweight prompt encoder, and a mask decoder. The segmentation output is conditioned on various user-provided prompts such as points, boxes, or masks. Trained on over one billion masks, SAM demonstrates strong generalization capabilities across domains, without requiring task-specific fine-tuning. Recent studies have focused on creating lightweight versions of SAM [43], such as Mobile SAM [44] and FastSAM [45], which aim to reduce computational complexity and improve inference speed. These adaptations make SAM more suitable for real-time applications and resource-constrained environments, while still maintain strong performance across various tasks.

However, directly applying SAM to RS images remains challenging due to the distinct characteristics of RS data. Compared with natural images, RS imagery typically covers large areas, exhibits complex textures, and contains many small and densely distributed objects [46]. In addition, spectral variation, sensor diversity, and high-spatial resolution hinder the direct use of SAM in RS scenarios. To address these limitations, recent studies have explored adaptations of SAM for RS-oriented segmentation tasks. For instance, RSPrompter [47] proposes fine-tuning methods and architectural modifications to better handle the extraction of buildings and roads from RS data.

Several recent studies have applied SAM to change detection in RS. For example, SAM-CD [48] utilized the visual encoder of SAM to extract bitemporal change features and introduced a convolutional adapter to aggregate feature information tailored for change detection. Similarly, Mei et al. [49] proposed SCD-SAM, which integrates the visual encoder from Mobile SAM and a convolution-based adaptor to improve the change detection performance on high-resolution RS images, achieving better accuracy than existing methods. Li et al. [50] leveraged the SAM encoder with a Fourier adapter and edge-constrained warping to enhance building change detection.

Despite the success of SAM in segmentation tasks, it has not yet been adapted for HSCD, particularly in the context of spatial misalignment. Current applications of SAM are primarily focused on segmentation, with limited exploration about the affine misregistration HSCD.

III. Proposed Method

A. Method Overview

As illustrated in Fig. 1, we propose a framework for HSCD under misregistration. The core of our framework establishes an intrinsic connection between registration and change detection through object-level priors. Leveraging the structural segmentation provided by SAM, registration operates jointly with detection within a shared semantic space. Consequently, the two tasks are unified under object-level consistency constraints, allowing the detection process to naturally absorb structural information from registration and thereby enhancing robustness in scenarios with spatial misalignment.

We denote the two HSIs acquired at times T_1 and T_2 as $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{H\times W\times C}$. To highlight the role of structural priors in our framework, the overall relationship among the key components can be expressed as follows:

$$\underbrace{\mathcal{R}\left(\mathbf{X}_1, \mathbf{X}_2\right)}_{\text{Registration}} \leftarrow \text{SAM}\left(\mathbf{X}_1, \mathbf{X}_2\right) \rightarrow \underbrace{\mathcal{F}_{\text{CD}}\left(\mathbf{X}_1, \mathbf{X}_2\right)}_{\text{Change Detection}} \tag{1}$$

where $\mathcal{R}$ denotes registration process based on object-level region correspondence, $\mathcal{F}_{\text{CD}}(\cdot)$ represents the change detection network, and SAM$(\cdot)$ represents the SAM.

We first convert each HSI into a three-channel pseudo-color projection to make it compatible with SAM. SAM then produces object-level segmentation masks, which provide stable region delineations for establishing ROI correspondences across time. By matching these ROIs and comparing their structural descriptors, we estimate an affine transform that brings corresponding object regions into spatial agreement, a structure-level alignment. The aligned masks serve as the basis for propagating sparse annotations within consistent object regions, thereby producing denser, object-aware supervision. The original hyperspectral pair together with the augmented labels are used to train a spatio-spectral-temporal change detection model that fuses convolutional local operators with Mamba-based sequential modules to capture both local spatial–spectral cues and long-range dependencies, yielding pixelwise change maps.

In the following, we provide a detailed description of the proposed method, which consists of four main components: 1) SAM-guided object-level registration; 2) region-based label propagation via mask intersection; 3) SST-MCDNet; and 4) loss function.

B. SAM-Guided Object-Level Registration

The inaccurate alignment between multitemporal HSIs introduces pixel mismatches that degrade the change detection

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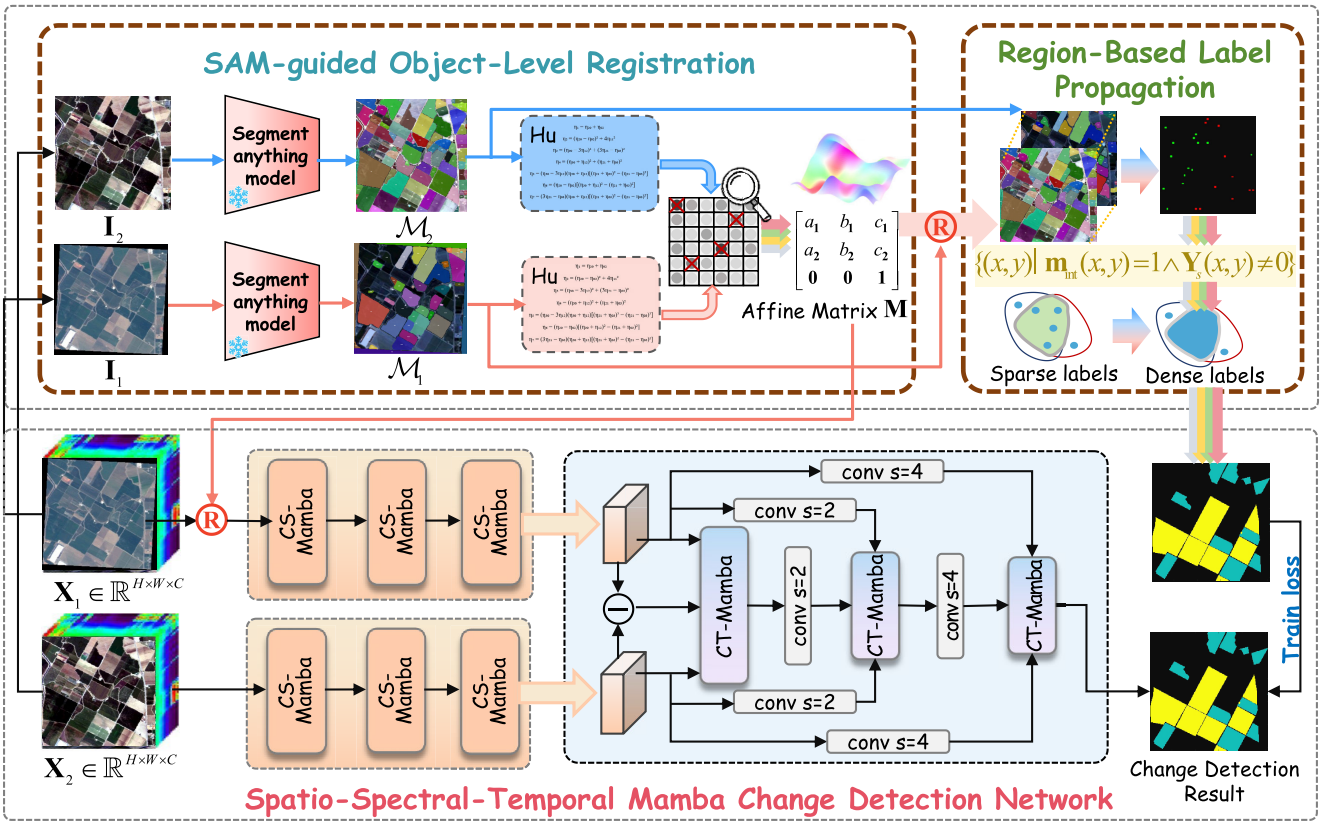


Fig. 1. Overview of the proposed SAM-Mamba guided framework for affine-dominated misalignment HSCD. The framework consists of SAM-guided object-level registration and SST-MCDNet. SAM-guided object-level registration is used to generate bitemporal HSIs with structure-consistent alignment, while SST-MCDNet performs the pixel-level change detection by jointly modeling spatial, spectral, and temporal dependencies.

performance. Instead of relying on pixel-level texture or intensity features, which are often unreliable in complex scenes, we exploit object-level priors to achieve the robust registration. Specifically, we leverage segmentation masks from the SAM to identify semantically consistent regions across time. These masks provide a stable structure-aware representation that enables us to extract reliable region correspondences and estimate an affine transformation for spatial alignment. The specific implementation process of the above SAM-guided object-level registration is illustrated in Fig. 2.

To adapt the hyperspectral data to the input requirements of SAM, we select three representative spectral bands from the visible wavelength range whose center wavelengths are closest to the red, green, and blue spectra, respectively, and map them to the R, G, and B channels to construct pseudo-color images $\mathbf{I}_1$ and $\mathbf{I}_2$

$$\mathbf{I}_i = \mathrm{Vis}\left(\mathbf{X}_i\right), \quad i = 1, 2 \tag{2}$$

where $\mathrm{Vis}(\cdot)$ denotes the operation for extracting pseudo-color representations from hyperspectral data. The spectral compression has a negligible impact on the subsequent registration process, since SAM primarily relies on spatial structures and geometric cues to extract object masks and estimate the affine transformation, rather than fine-grained spectral information. Next, we apply the SAM to obtain segmentation masks $\mathcal{M}_1$ and $\mathcal{M}_2$ from the pseudo-color images

$$\mathcal{M}_i = \mathrm{SAM}\left(\mathbf{I}_i\right) = \left\{\mathbf{m}_i^k\right\}_{k=1}^{K_i}, \quad \mathbf{m}_i^k \in \{0,1\}^{H\times W}, \quad i = 1, 2 \tag{3}$$

where each binary mask $\mathbf{m}_i^k$ corresponds to a spatially coherent region, and K_i is the total number of such regions in image $\mathbf{I}_i$. Although SAM does not provide explicit semantic labels, the generated masks effectively capture consistent object-like regions and delineate their boundaries, serving as valuable structural cues for alignment.

To quantitatively describe the shape characteristics of each segmented region, we compute the seven Hu moments $\mathbf{h}_i^k \in \mathbb{R}^7$ for every mask $\mathbf{m}_i^k$

$$\mathbf{h}_i^k = \mathrm{HuMoments}\left(\mathbf{m}_i^k\right). \tag{4}$$

We then perform region matching by computing the pairwise Euclidean distances d_{uv} between Hu moment vectors from the two image sets

$$d_{uv} = \|\mathbf{h}_1^u - \mathbf{h}_2^v\|_2 \tag{5}$$

where a smaller distance indicates higher shape similarity. Let $\{d_{uv}\}$ be the complete distance set across all region pairs (u, v). By ranking these distances in ascending order, we obtain the top-N pairs of regions with the highest similarity, denoted as follows:

$$\mathcal{P} = \{(u_n, v_n)\}_{n=1}^{N} \tag{6}$$

where (u_n, v_n) indexes the nth closest pair according to d_{uv}. These correspondences reflect candidate region matches that are likely to represent the same or structurally similar objects across the two images.

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For each candidate pair (u_n, v_n), we compute the centroid coordinates of the corresponding regions

$$\mathbf{c}_i^{u v_n} = \text{centroid}\left(\mathbf{m}_i^{u v_n}\right), \quad i = 1, 2 \tag{7}$$

where $\mathbf{c}_i^{u v_n}$ serve as representative points for spatial alignment. The set of centroid pairs extracted from $\mathcal{P}$ provides the basis for geometric transformation estimation.

To estimate the affine transformation matrix $\mathbf{A}$ that best aligns the centroids from $\mathbf{X}_2$ to $\mathbf{X}_1$, we employ the RANSAC algorithm [51], which robustly fits the model while mitigating the influence of outliers caused by segmentation errors or changed regions. The optimization objective is formulated as follows:

$$\mathbf{M} = \arg\min_{\mathbf{M}} \sum_{n=1}^{N} \left\| \mathbf{c}_1^{k_n} - \mathbf{A}\tilde{c}_2^{l_n} \right\|_2^2. \tag{8}$$

The affine matrix $\mathbf{M}$ encapsulates translation, rotation, scaling, and shearing, providing a flexible yet interpretable mapping.

The resulting affine transformation matrix $\mathbf{M}$ is then applied to obtain $\mathbf{X}_2^{\text{warm}}$ and its associated segmentation masks, aligning them to the coordinate system of $\mathbf{X}_1$. This alignment step significantly reduces spatial misregistration, thereby improving the consistency of subsequent change detection processing.

Overall, the proposed method effectively integrates SAM-based structure-aware segmentation with invariant shape descriptors and robust geometric fitting. Although Hu moments can be sensitive to noise in low-resolution data, our framework ensures stability by leveraging the regularization effect of SAM masks and relying on relative descriptor ranking rather than absolute values. Furthermore, the use of RANSAC on region centroids effectively filters out potential outliers. This approach achieves reliable alignment without relying on dense pixel-level correspondences or manually labeled registration data and enhances the robustness of HSCD under challenging spatial misalignment conditions.

C. Region-Based Label Propagation via Mask Intersection

To incorporate object-level structural priors into HSCD, we perform label augmentation guided by SAM-generated segmentation masks. These masks delineate semantically coherent regions across time, allowing us to propagate sparse annotations to corresponding areas within each object, as illustrated in Fig. 1. By extending sparse change labels to mask-overlapping regions, we effectively encode object-level consistency into the learning process.

Specifically, after obtaining the affine transformation matrix $\mathbf{A}$, we apply it to the segmentation masks $\mathcal{M}_2$ generated by SAM, yielding the aligned mask set $\mathcal{M}_2'$. This transformation brings the regions in the second time-phase image into spatial correspondence with those in the first image. We then utilize these aligned masks to propagate sparse ground-truth labels by leveraging region-level consistency. We design an SAM-guided object-level registration (SOR) module that aligns bitemporal images by leveraging region-level masks extracted from SAM.

Let $\mathbf{Y}_s \in \{0, 1, 2\}^{H \times W}$ be the sparse label map, where 0 indicates unlabeled pixels, 1 denotes change, and 2 denotes no change. For each pair of corresponding masks $(\mathbf{m}_1^u, \mathbf{m}_2'^v)$, we compute the intersection mask

$$\mathbf{m}_{\text{int}} = \mathbf{m}_1^u \wedge \mathbf{m}_2'^v \tag{9}$$

and extract the set of labeled pixels within this intersection

$$\Omega = \{(x, y) \mid \mathbf{m}_{\text{int}}(x, y) = 1 \wedge \mathbf{Y}_s(x, y) \neq 0\}. \tag{10}$$

If all labels in Ω share the same value $m \in \{1, 2\}$, we consider the region to be label-consistent and assign m to the entire intersection area

$$\mathbf{Y}_{\text{aug}}(x, y) = m, \quad \forall (x, y) \in \mathbf{m}_{\text{int}}. \tag{11}$$

The expansion process iterates over all intersecting ROI pairs to generate a pseudo-dense label map $\mathbf{Y}_{\text{aug}}$, which contains spatially coherent supervision signals. Compared with the pixelwise interpolation or manual annotation, this strategy leverages the structural regularity provided by SAM masks and the alignment from the affine transformation to enhance label coverage in a reliable and interpretable manner.

D. Spatio-Spectral-Temporal Mamba Change Detection Network

The accurate results of HSCD demand comprehensive modeling of spatial, spectral, and temporal characteristics. Therefore, we propose the SST-MCDNet, which effectively integrates convolutional operations with the Mamba architecture to jointly capture local spatial–spectral features and long-range dependencies. Specifically, the network employs a dual-branch encoder where each branch independently processes one temporal HSI through a series of convolutional layers and spectral–spatial Mamba blocks, extracting rich hierarchical features. These encoded features are then fed into a CTFD, which models bidirectional temporal dependencies and center-guided difference attention to emphasize meaningful changes between the two time points. By fusing convolutional local cues with Mamba-based sequential modeling, SST-MCDNet is capable of robustly distinguishing subtle spectral and spatial variations over time, enabling the precise pixel-level change detection in challenging hyperspectral scenarios.

In the proposed framework, the two temporal HSIs are independently processed by a dual-branch encoder. Each branch employs a combination of convolutional layers and spectral–spatial Mamba blocks to extract hierarchical representations that capture both local spatial–spectral patterns and long-range dependencies. The outputs of the dual branches are then denoted as follows:

$$(\mathbf{F}_1, \mathbf{F}_2) = \text{Encoder}\left(\mathbf{X}_1, \mathbf{X}_2^{\text{warm}}\right) \tag{12}$$

where $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{H \times W \times D}$ are the extracted spatio-spectral features from $\mathbf{X}_1$ and $\mathbf{X}_2$, respectively.

These features are then fed into the CTFD to estimate the final change map

$$\mathbf{Y} = \text{Decoder}(\mathbf{F}_1, \mathbf{F}_2) \tag{13}$$

where $\mathbf{Y} \in \mathbb{R}^{H \times W}$ denotes the predicted change probability map.

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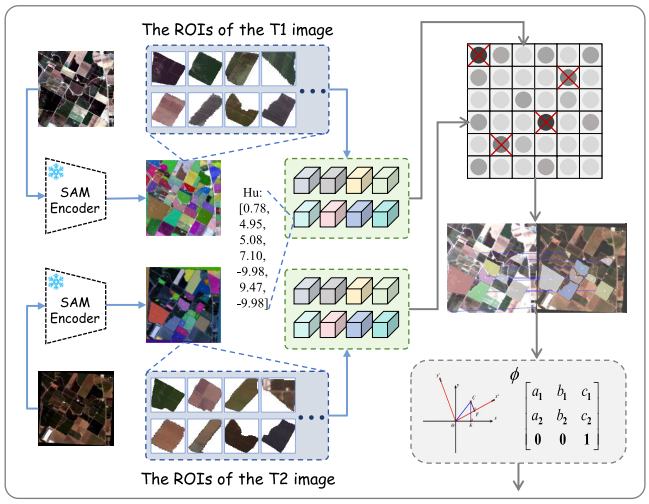


Fig. 2. SAM-guided object-level registration framework.

1) Spectral–Spatial Contextual Encoder: We first describe the encoder module, which aims to capture spatial structures and spectral correlations from each input HSI, providing rich representations for subsequent change decoding. To this end, we propose a dual-branch encoder architecture in which each temporal input is processed independently through a sequence of hybrid convolutional and spectral–spatial Mamba blocks, as illustrated in Fig. 3.

The encoder adopts a dual-branch architecture, where each branch independently processes one temporal HSI ($\mathbf{X}_1$ or $\mathbf{X}_2$). Each branch is composed of a stack of N spectral–spatial convolutional Mamba blocks (SSCM-Blocks), which are designed to jointly model local spatial textures and global spectral dependencies. The hierarchical stacking enables the encoder to extract multilevel features that preserve fine-grained spatial details while enhancing the spectral continuity and semantic consistency. Formally, for each time $t \in \{1, 2\}$, the encoder representation can be expressed as follows:

$$\mathbf{F}_t = \text{Encoder}(\mathbf{X}_t) = \text{SSCM}_N(\ldots \text{SSCM}_1(\mathbf{X}_t)) \tag{14}$$

where $\text{SSCM}_i(\cdot)$ denotes the ith SSCM-Block in the stack and $\mathbf{F}_t$ is the final encoded feature map corresponding to temporal input $\mathbf{X}_t$.

The SSCM-Block consists of two parallel branches: a spatial modeling branch and a spectral modeling branch. In the spatial branch of the SSCE module, we aim to extract long-range contextual dependencies along dominant spatial directions while preserving local structural details. Given an input tensor $\mathbf{X}$, we first reshape it into four 1-D sequences along canonical axes: horizontal ($\rightarrow$), reverse-horizontal ($\leftarrow$), vertical ($\downarrow$), and reverse-vertical ($\uparrow$). These directional sequences are denoted by $\{\mathbf{x}_t^{\rightarrow}\}_{t=1}^{T}$ and $\{\mathbf{x}_t^{\leftarrow}\}_{t=1}^{T}$, where $L = H \times W$.

Each sequence is modeled by a unified 1-D SSM to capture global dependencies across spatial positions. The SSM is defined as follows:

$$\mathbf{h}_t = \mathbf{A}\mathbf{h}_{t-1} + \mathbf{B}\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t \tag{15}$$

where $\mathbf{x}_t \in \mathbb{R}^d$ is the input token at time t, $\mathbf{h}_t \in \mathbb{R}^{d_h}$ is the hidden state, and $\mathbf{y}_t \in \mathbb{R}^d$ is the output. The matrices $\mathbf{A} \in \mathbb{R}^{d_h \times d_h}$, $\mathbf{B} \in \mathbb{R}^{d_h \times d}$, and $\mathbf{C} \in \mathbb{R}^{d \times d_h}$ are shared across all directions and learned during training.

In this formulation, $\mathbf{A}$ governs the temporal memory dynamics, enabling accumulation of context from previous tokens. $\mathbf{B}$ injects new directional observations into the hidden state, while $\mathbf{C}$ projects the hidden dynamics back into the output space. This linear recurrence design allows efficient modeling of long-range dependencies with linear time and memory complexity.

The directional outputs are reshaped back into spatial feature maps: $\mathbf{S}^{\rightarrow}$, $\mathbf{S}^{\leftarrow}$, $\mathbf{S}^{\downarrow}$, and $\mathbf{S}^{\uparrow}$. The averaged result $\mathbf{F}_{\text{spa}}$ can be formulated as follows:

$$\mathbf{F}_{\text{spa}} = \frac{1}{4}\left(\mathbf{S}^{\rightarrow} + \mathbf{S}^{\leftarrow} + \mathbf{S}^{\downarrow} + \mathbf{S}^{\uparrow}\right). \tag{16}$$

In parallel with the directional modeling, a local spatial feature map $\mathbf{F}_{\text{loc}} \in \mathbb{R}^{H \times W \times d}$ is extracted from the input $\mathbf{X}$ using a two-layer convolutional block with ReLU activation and batch normalization. The process is formulated as follows:

$$\mathbf{F}_{\text{loc}} = \text{ConvBlocks}(\mathbf{X}) \tag{17}$$

where ConvBlocks$(\cdot)$ denotes a stack of convolutional layers followed by ReLU activation and batch normalization. The resulting local features $\mathbf{F}_{\text{loc}}$ are subsequently used both for spatial feature fusion and as the basis for the attention mechanism that enhances the structural discrimination.

To enhance structural discrimination, we compute both average and max pooling along the channel dimension of $\mathbf{F}_{\text{loc}}$, concatenate the results, and apply a convolutional layer followed by softmax activation to produce an attention map

$$\mathbf{A}_{\text{spa}} = \text{Softmax}\left(\text{Conv}\left(\text{Pool}_{\text{avg+max}}(\mathbf{F}_{\text{loc}})\right)\right) \tag{18}$$

where $\text{Pool}_{\text{avg+max}}(\cdot)$ denotes the channelwise pooling and concatenation. The resulting $\mathbf{A}_{\text{spa}} \in \mathbb{R}^{H \times W \times 1}$ highlights structurally salient regions, which are used to reweight the aggregated spatial representation

$$\mathbf{F}'_{\text{spa}} = \left(\mathbf{F}_{\text{spa}} + \mathbf{F}_{\text{loc}}\right) \odot \mathbf{A}_{\text{spa}} \tag{19}$$

where $\odot$ denotes the elementwise multiplication. This spatially aware feature $\mathbf{F}'_{\text{spa}}$ retains both global contextual information and local structural cues, and serves as one of the dual-branch outputs in the SSCE module.

In parallel with the spatial branch, the SSCE module includes a spectral modeling branch to capture long-range dependencies along the spectral dimension. Given the input tensor $\mathbf{X}$, we reshape it into spectral sequences at each spatial position, yielding $\{\mathbf{x}_t\}_{t=1}^{C}$. These spectral vectors are then modeled using two directional Mamba modules: one in the forward (channelwise) order and the other in the reverse order.

Each directional Mamba follows a simplified SSM formulation. For a sequence $\{\mathbf{x}_t\}_{t=1}^{C}$, the model computes

$$\mathbf{h}_t = \mathbf{A}_s\mathbf{h}_{t-1} + \mathbf{B}_s\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_s\mathbf{h}_t \tag{20}$$

where $\mathbf{A}_s$, $\mathbf{B}_s$, and $\mathbf{C}_s$ are learnable matrices, and $\mathbf{y}_t$ denotes the spectral feature at step t. The outputs from both directions are aggregated and reshaped to form a spectral feature map

$$\mathbf{F}_{\text{spe}} = \frac{1}{2}\left(\overrightarrow{\mathbf{S}} + \overleftarrow{\mathbf{S}}\right). \tag{21}$$

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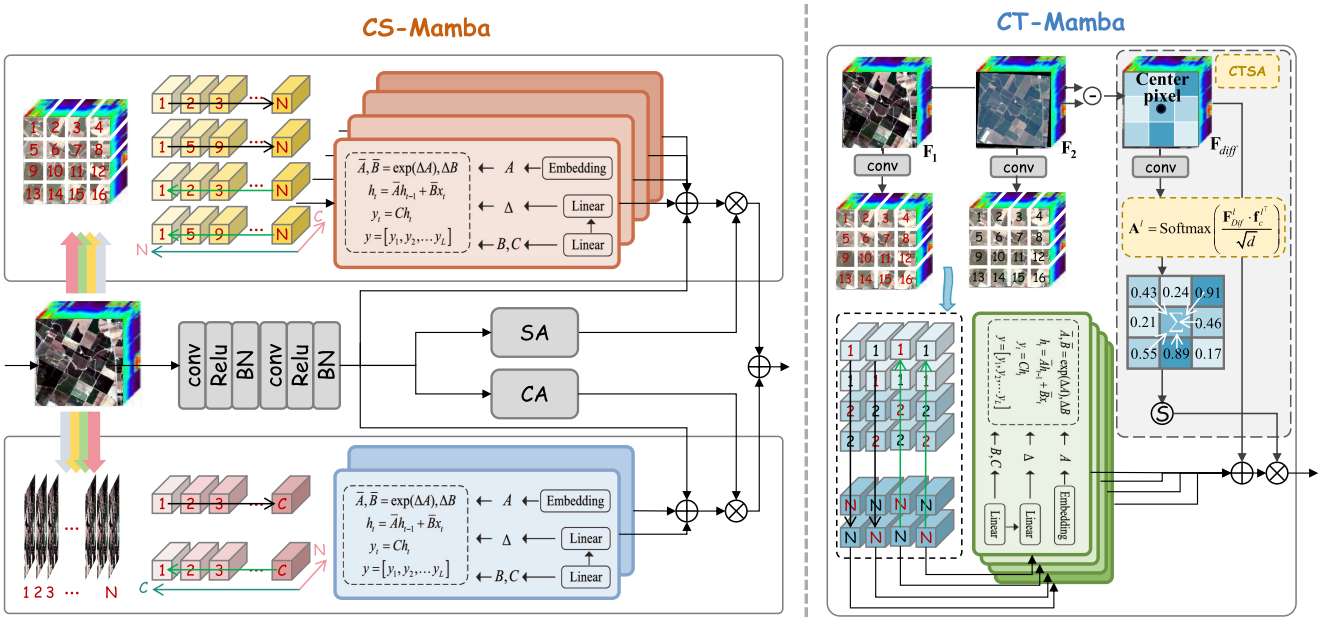


Fig. 3. Detailed architecture of the spatio-spectral-temporal encoder and CTFD.

To enhance the discriminative capability of spectral features, we incorporate a channel attention mechanism. Specifically, average and max pooling are performed on $\mathbf{F}_{\text{loc}}$ along spatial dimensions, and the resulting descriptors are concatenated and processed by a convolutional layer followed by a sigmoid activation

$$\mathbf{A}_{\text{spe}} = \sigma\left(\text{Conv}\left(\text{Pool}_{\text{avg+max}}\left(\mathbf{F}_{\text{loc}}\right)\right)\right) \tag{22}$$

where $\mathbf{A}_{\text{spe}} \in \mathbb{R}^{1\times 1\times d}$ is broadcasted and used to reweight the spectral representation

$$\mathbf{F}'_{\text{spe}} = \left(\mathbf{F}_{\text{spe}} + \mathbf{F}_{\text{loc}}\right) \odot \mathbf{A}_{\text{spe}}. \tag{23}$$

Finally, the attention-enhanced spectral feature $\mathbf{F}'_{\text{spe}}$ is added to the spatial output $\mathbf{F}'_{\text{spa}}$ to form the final encoded representation

$$\mathbf{F}_{\text{enc}} = \mathbf{F}'_{\text{spa}} + \mathbf{F}'_{\text{spe}}. \tag{24}$$

2) Center-Aware Temporal Difference Decoder: To model temporal dynamics and extract change-relevant features between dual-temporal hyperspectral observations, we propose the CTFD. The decoder is composed of a series of center-aware temporal Mamba blocks (CTMBs), each jointly modeling temporal context and localized difference attention, as illustrated in Fig. 3.

The CTMB simultaneously ingests three types of features from the previous layer: the individual temporal features from T_1 and T_2, as well as their difference representation. These features are jointly refined through sequential temporal modeling and attention-driven fusion. Formally, the decoding process across layers is defined as follows:

$$\mathbf{F}_{\Delta}^{(l+1)} = \text{CTMB}\left(\mathbf{F}_{D1}^{(l)}, \mathbf{F}_{D2}^{(l)}, \mathbf{F}_{\Delta}^{(l)}\right), \quad l = 0, 1, \ldots, N-1 \tag{25}$$

where CTMB$(\cdot)$ denotes CTMB, the two time-specific features $\mathbf{F}_{D1}^{(l)}$ and $\mathbf{F}_{D2}^{(l)}$ are first updated via shallow convolutional layers to extract localized information. The initial inputs are set as $\mathbf{F}_{D1}^{(0)} = \mathbf{F}_{\text{enc}}^{1}$ and $\mathbf{F}_{D2}^{(0)} = \mathbf{F}_{\text{enc}}^{2}$, corresponding to the encoder outputs of $\mathbf{X}_1$ and $\mathbf{X}_2$. These updated features, along with the different representation $\mathbf{F}\Delta^{(l)}$ from the previous layer, are then processed by the CTMB to obtain the next-level difference feature.

At lth layer of the CTMB, the updated features $\mathbf{F}_{D1}^{(l)}$ and $\mathbf{F}_{D2}^{(l)}$ are first spatially divided into N nonoverlapping patches and then transformed into sequential tokens. Let $\mathbf{f}_1^{(i)}$ and $\mathbf{f}_2^{(i)}$ be the ith tokens of the two time steps ($i = 1, \ldots, N$). These tokens are arranged into four interleaved sequences to model bidirectional cross-temporal dependencies

$$\begin{aligned}
\mathcal{S}_1 &= \left[\mathbf{f}_1^{(1)}, \mathbf{f}_2^{(1)}, \ldots, \mathbf{f}_1^{(N)}, \mathbf{f}_2^{(N)}\right] \\
\mathcal{S}_2 &= \left[\mathbf{f}_2^{(1)}, \mathbf{f}_1^{(1)}, \ldots, \mathbf{f}_2^{(N)}, \mathbf{f}_1^{(N)}\right] \\
\mathcal{S}_3 &= \left[\mathbf{f}_1^{(N)}, \mathbf{f}_2^{(N)}, \ldots, \mathbf{f}_1^{(1)}, \mathbf{f}_2^{(1)}\right] \\
\mathcal{S}_4 &= \left[\mathbf{f}_2^{(N)}, \mathbf{f}_1^{(N)}, \ldots, \mathbf{f}_2^{(1)}, \mathbf{f}_1^{(1)}\right].
\end{aligned} \tag{26}$$

Each interleaved sequence $\mathcal{S}_i^l$ is individually fed into an SSM to capture long-range cross-temporal dependencies under different directional orders. The outputs from the four directions are then aggregated to obtain a unified temporal representation

$$\mathbf{F}_{\text{seq}}^{l} = \sum_{i=1}^{4} \text{SSM}\left(\mathcal{S}_i^l\right) \tag{27}$$

where $\mathbf{F}_{\text{seq}}^{l}$ denotes the temporal representation and SSM$(\cdot)$ denotes the SSM.

HSCD adopts a pixelwise computation paradigm based on patches, where each patch provides spatial and neighborhood information to support the change detection of its central pixel. However, nonhomogeneous information within the patch (i.e., pixels belonging to different change categories) may lead to misjudgment of the central pixel's category. To address

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this issue, the center difference similarity attention (CPSA) module is designed to amplify the contribution of pixels aligned with the central pixel's change tendency and suppress nonhomogeneous or ambiguous pixels, ensuring the patch's effective information is fully utilized without interference from irrelevant data, thereby accurately determining the true change state of the central pixel.

To obtain robust local representations before attention computation, the input difference feature $\mathbf{F}_{\Delta}^{(l)}$ is first passed through a shallow convolutional stack

$$\mathbf{F}_{\mathrm{Dif}}^{l} = \mathrm{ConvBlocks}\left(\mathbf{F}_{\Delta}^{(l)}\right). \tag{28}$$

To focus on the most relevant local differences for each center pixel, we adopt a scaled dot-product attention mechanism. Specifically, given the difference feature patch $\mathbf{F}_{\mathrm{Dif}}^{l} \in \mathbb{R}^{P\times d}$ and the center feature vector $\mathbf{f}_{c}^{l} \in \mathbb{R}^{d}$, the attention weights $\mathbf{A}_{\mathrm{c}}^{l}$ are computed as follows:

$$\mathbf{A}_{\mathrm{c}}^{l} = \mathrm{Softmax}\left(\frac{\mathbf{F}_{\mathrm{Dif}}^{l}\cdot \mathbf{f}_{c}^{l\top}}{\sqrt{d}}\right) \tag{29}$$

where d is the feature dimension and the dot product is scaled by $\sqrt{d}$ to maintain numerical stability. The attention-enhanced feature is then obtained via weighted aggregation.

Algorithm 1 Algorithm for the Proposed SAM-Guided SST-Mamba Framework

Input: Bi-temporal HSIs $\mathbf{X}_1$ (T1), $\mathbf{X}_2$ (T2);
Sparse label map $\mathbf{Y}_s$;
CTMB layers N; training epochs Max_epoch; learning rate η;
Initialize model parameters Θ;
Output: Pixel-wise change detection map $\mathbf{Y}_{\mathrm{pre}}$;
Step 1: Generate pseudo-color images for SAM
Generate pseudo-color images from visible bands of $\mathbf{X}_1$, $\mathbf{X}_2$;
Step 2: SAM-guided object-level registration
Get segmentation masks via SAM, estimate affine matrix (Hu moments + RANSAC), warp $\mathbf{X}_2$ to $\mathbf{X}_2^{\mathrm{warp}}$;
Step 3: Region-based label propagation
Augment sparse labels $\mathbf{Y}_s$ to dense map $\mathbf{Y}_{\mathrm{aug}}$ via region intersection matching;
Step 4: Model training (SST-SCDNet)
$e \leftarrow 1$;
while $e \le$ Max_epoch **do**
Predict $\mathbf{Y}_{\mathrm{pre}}^{(e)}$ and calculate BCE loss $\mathcal{L}_{\mathrm{BCE}}$;
Optimize Θ via backpropagation and update optimizer;
$e \leftarrow e + 1$;
end while
Step 5: Final inference
Generate $\mathbf{Y}_{pre}$;
return $\mathbf{Y}_{pre}$;

The attended features $\mathbf{F}_{\mathrm{att}}$ are used to reweight the fused dual-time features. Specifically, the final decoded representation is

$$\mathbf{F}_{\Delta}^{l+1} = \left(\mathbf{F}_{\mathrm{Dif}}^{l} + \mathbf{F}_{\mathrm{seq}}^{l}\right) \odot \mathbf{A}_{\mathrm{c}}^{l} \tag{30}$$

where $\odot$ denotes the elementwise multiplication. The decoded features $\mathbf{F}_{\Delta}^{N}$ are then flattened and passed through two fully connected layers, followed by a softmax classifier to predict pixelwise change probabilities

$$\mathbf{Y}_{\mathrm{pre}} = \mathrm{Softmax}\left(\mathrm{FC}_2\left(\mathrm{ReLU}\left(\mathrm{FC}_1\left(\mathbf{F}_{\Delta}^{N}\right)\right)\right)\right) \tag{31}$$

where $\mathbf{Y}_{\mathrm{pre}}$ is the change probabilities, $\mathrm{FC}_i(\cdot)$ denotes the ith fully connected layer, and $\mathrm{ReLU}(\cdot)$ denote the ReLU activation function. The detailed algorithm is shown in Algorithm 1.

E. Loss Function

The training supervision is derived from the pseudo-labels (PLs) generated in Section III-C, where sparse manual annotations are propagated within object-level regions using SAM-based segmentation and registration. These extended labels provide more comprehensive supervision while maintaining structural consistency. To ensure the reliable optimization, we compute the binary cross-entropy loss only over pixels with valid labels

$$\mathcal{L}_{\mathrm{BCE}} = -\left(\mathbf{Y}_{\mathrm{aug}}\log\left(\mathbf{Y}_{\mathrm{pre}}\right) + \left(1-\mathbf{Y}_{\mathrm{aug}}\right)\log\left(1-\mathbf{Y}_{\mathrm{pre}}\right)\right) \tag{32}$$

where $\mathcal{L}_{\mathrm{BCE}}$ denotes the binary cross-entropy loss.

IV. Experimental Results and Analysis

To comprehensively evaluate the performance of the proposed SST-MCDNet and its supporting modules, we conduct extensive experiments on benchmark HSCD datasets. This section is organized into four subsections. First, we describe three datasets used for evaluation. Second, we detail the experimental settings, including data preprocessing, implementation details, training configurations, and the baseline methods used for comparison. Third, we present quantitative and qualitative results to demonstrate the superiority of our method over existing approaches. Finally, we conduct ablation studies to assess the effectiveness of each proposed module, including the object-level registration, label expansion, and SST-MCDNet.

A. Data Description

To comprehensively evaluate the effectiveness of the proposed SST-MCDNet, we conduct experiments on three representative HSCD datasets, as shown in Fig 4: *Barbara*, *Bay Area*, and *Hermiston*.

1) *Barbara:* It was acquired by the AVIRIS sensor over Santa Barbara, CA, USA, in 2013 and 2014. It contains two HSIs with a spatial resolution of 20 m and a spectral coverage from 0.4 to 2.5 μm at 10-nm intervals, totaling 224 spectral bands. The image size is 984×740 pixels, mainly covering urban areas with significant vegetation changes.
2) *Bay Area:* It was collected in 2013 and 2014 by the AVIRIS sensor and covers the Patterson region in California. It has a spatial resolution of 20 m, a spectral range of 0.4–2.5 μm with 224 bands, and an image size of 600×500 pixels. The scenes involve both built-up and natural areas.

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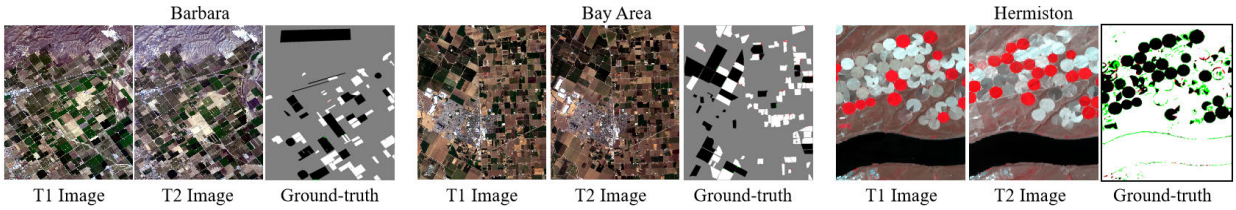


Fig. 4. Subjective visualization of pseudo-color image pairs and change detection labels from three datasets.

TABLE I

OBJECTIVE PERFORMANCE OF SDCO-MDDN AND OTHER COMPETING METHODS FOR BARBARA UNDER ALIGNED AND AFFINE-TRANSFORMED. BOLD INDICATES THE HIGHEST VALUE, AND UNDERLINE INDICATES THE SECOND HIGHEST VALUE

BAR	Method	CVA	RECNN	DADSSFF	DIEFEN	MSDFFN	MFCEN	Proposed
Aligned	OA	0.8716	0.9820	0.9797	0.9757	0.9884	0.9931	**0.9947**
	Kappa	0.7318	0.9622	0.9574	0.9494	0.9757	0.9856	**0.9888**
	F1	0.8383	0.9770	0.9740	0.9698	0.9853	0.9913	**0.9932**
Affine-transformed	OA	0.8802	0.9813	0.9806	0.9827	0.9801	0.9880	**0.9910**
	Kappa	0.7485	0.9608	0.9591	0.9640	0.9585	0.9747	**0.9812**
	F1	0.8468	0.9762	0.9749	0.9784	0.9751	0.9846	**0.9886**

3) *Hermiston:* It includes HSIs acquired by the EO-1 sensor over Hermiston, OR, USA, on May 1, 2004 and May 8, 2007. It covers wavelengths from 0.4 to 2.5 μm with a spectral resolution of 10 nm. The spatial resolution is approximately 30 m, and each image contains 242 spectral bands, among which 154 are selected for use. The region includes agricultural land and river surfaces.

B. Experimental Setup

To rigorously assess the effectiveness and robustness of the proposed framework, all experiments are conducted under both geometrically aligned and deliberately misaligned conditions. For the misaligned setting, we synthetically introduce geometric distortions to the second temporal image using affine transformations. Specifically, rotation angles are randomly sampled from the range $[-5°, 5°]$, while horizontal and vertical translations are uniformly drawn from $[-5, 5]$ pixels. To minimize spectral distortion during geometric manipulation, nearest-neighbor interpolation is employed throughout the transformation process. In all experiments, training is performed under a sparse supervision regime, where only 1% of the total pixels are randomly sampled from both the change and no-change classes to serve as labeled data. This setting closely reflects real-world scenarios where densely labeled hyperspectral datasets are rarely available.

Model training is conducted on a workstation equipped with an Intel Core i9-10900X CPU and an NVIDIA RTX 3090 GPU with 24 GB of memory. The models are optimized using the Adam optimizer, with the initial learning rate set to 0.01 and reduced by a factor of 2 every 50 epochs. Unless otherwise specified, all experiments adopt the same training protocol to ensure fair comparison.

C. Comparison Methods and Evaluation Metrics

To comprehensively evaluate the effectiveness of our proposed approach, we compare it against six methods, including one classical algorithm CVA [10] and five state-of-the-art deep learning-based change detection models: RECNN [20], DADSSFF [18], DIEFEN [22], MFCEN [52], and MSDFFN [13]. To ensure a fair comparison, all methods adopt identical data splits and training settings. In addition, considering the presence of spatial misalignment in the bitemporal data, a standard feature-based registration step based on SURF [53] is uniformly applied as a preprocessing stage for all comparative methods, so as to guarantee spatial consistency of the inputs. Performance is quantitatively evaluated using three widely recognized metrics: overall accuracy (OA), Kappa coefficient, and the $F1$-score. OA reflects the global classification performance across all pixels, Kappa accounts for random agreement and provides a more robust measure, while $F1$ balances precision and recall for the change class, particularly important in scenarios with imbalanced data distributions.

D. Quantitative and Qualitative Results

1) Quantitative Results: As shown in Tables I–III, our method consistently achieves the highest performance across all three datasets under both aligned and affine-transformed settings. Across the aligned scenarios, it outperforms all baseline methods, including traditional CVA and state-of-the-art deep learning models such as MSDFFN and MFCEN, achieving near-perfect accuracy, high Kappa coefficients, and strong $F1$-scores. Under the affine-transformed scenarios, although all methods undergo uniform SURF-based registration preprocessing, most baselines still experience noticeable performance degradation due to residual spatial misalignment. In contrast, our method maintains robust performance with minimal loss, demonstrating strong tolerance to geometric distortions and effectively leveraging SAM-based semantic registration combined with label expansion.

2) Qualitative Results: Visual comparisons, as shown in Figs. 5–7, further corroborate the quantitative findings. The proposed method consistently produces clearer and more coherent change maps across all datasets, with well-preserved

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TABLE II

OBJECTIVE PERFORMANCE OF THE PROPOSED AND OTHER COMPETING METHODS FOR BAY AREA UNDER ALIGNED AND AFFINE-TRANSFORMED. BOLD INDICATES THE HIGHEST VALUE, AND UNDERLINE INDICATES THE SECOND HIGHEST VALUE

BAY	Method	CVA	RECNN	DADSSFF	DIEFEN	MSDFFN	MFCEN	Proposed
Aligned	OA	0.8517	0.9430	0.9758	0.9769	0.9794	<u>0.9816</u>	**0.9856**
	Kappa	0.7063	0.8858	0.9515	0.9535	0.9587	<u>0.9631</u>	**0.9712**
	F1	0.8462	0.9460	0.9770	0.9786	0.9806	<u>0.9827</u>	**0.9864**
Affine-transformed	OA	0.8397	0.9378	0.9654	<u>0.9758</u>	0.9743	0.9738	**0.9801**
	Kappa	0.6833	0.8751	0.9307	<u>0.9514</u>	0.9484	0.9472	**0.9599**
	F1	0.8301	0.9416	0.9669	<u>0.9776</u>	0.9757	0.9756	**0.9815**

TABLE III

OBJECTIVE PERFORMANCE OF THE PROPOSED AND OTHER COMPETING METHODS FOR HERMISTON UNDER ALIGNED AND AFFINE-TRANSFORMED. BOLD INDICATES THE HIGHEST VALUE, AND UNDERLINE INDICATES THE SECOND HIGHEST VALUE

Hermiston	Method	CVA	RECNN	DADSSFF	DIEFEN	MSDFFN	MFCEN	Proposed
Aligned	OA	0.9202	0.8906	0.9430	0.9358	<u>0.9464</u>	0.9416	**0.9661**
	Kappa	0.7419	0.6882	0.8289	0.8085	<u>0.8419</u>	0.8294	**0.9017**
	F1	0.7888	0.9293	0.8648	0.9592	<u>0.9658</u>	0.8667	**0.9782**
Affine-transformed	OA	0.9130	0.9097	0.9368	0.9253	0.9339	<u>0.9375</u>	**0.9582**
	Kappa	0.7205	0.7391	<u>0.8509</u>	0.7866	0.8104	0.8066	**0.8773**
	F1	0.9462	0.9425	0.9603	0.9517	0.9574	<u>0.9609</u>	**0.9733**

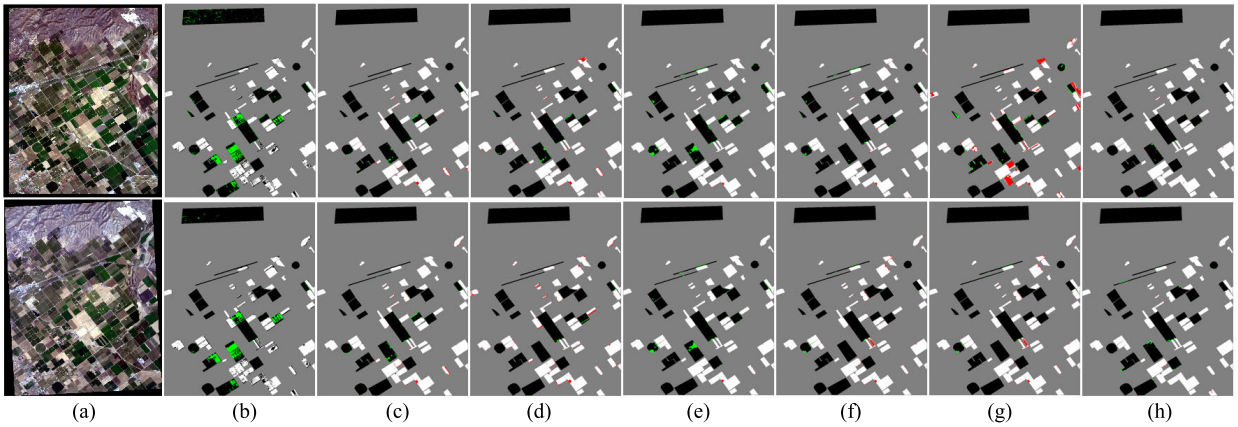


Fig. 5. Visualization of change detection results on the Barbara dataset: (top row) with aligned and (bottom row) with affine-transformed. (a) T2 images. (b) CVA. (c) RECNN. (d) DADSSFF. (e) DIEFEN. (f) MSDFFN. (g) MFCEN. (h) Proposed.

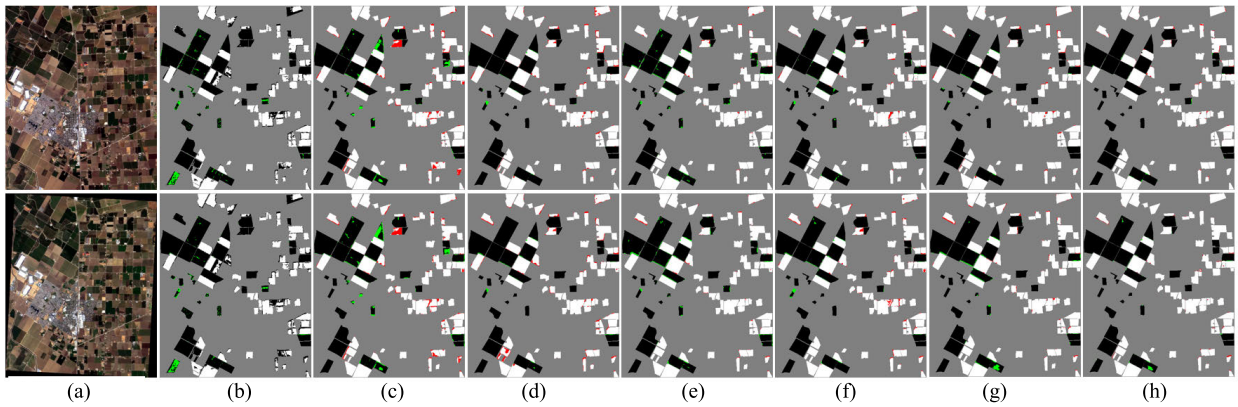


Fig. 6. Visualization of change detection results on the Bay Area dataset: (top row) with aligned and (bottom row) with affine-transformed. (a) T2 images. (b) CVA. (c) RECNN. (d) DADSSFF. (e) DIEFEN. (f) MSDFFN. (g) MFCEN. (h) Proposed.

object boundaries and fewer false positives and false negatives. In contrast, conventional methods, including traditional CVA as well as deep learning baselines, such as RECNN, DADSSFF, and MSDFFN, often generate noisy, blurry, or

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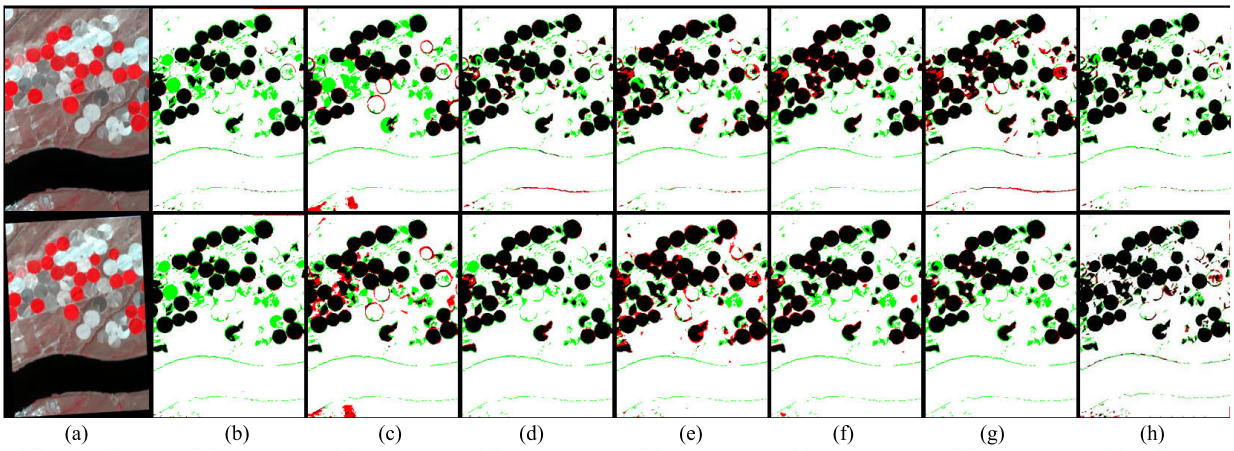


Fig. 7. Visualization of change detection results on the Hermiston dataset: (top row) with aligned and (bottom row) with affine-transformed. (a) T2 images. (b) CVA. (c) RECNN. (d) DADSSFF. (e) DIEFEN. (f) MSDFFN. (g) MFCEN. (h) Proposed.

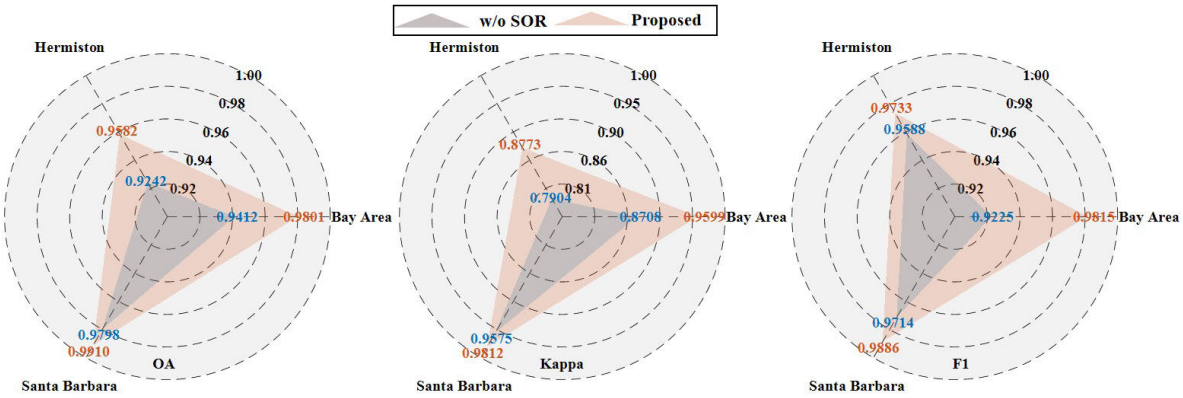


Fig. 8. Performance comparison of the proposed method and its variant which removes SOR on three datasets.

fragmented results when spatial inconsistencies are present. The superior visual quality of our results can be attributed to the integration of object-level priors, spatially invariant feature learning, and robust registration-aware label expansion, which together ensure precise and fine-grained change detection even under nonideal geometric conditions.

E. Ablation Study

To investigate the contribution of each core component in the proposed method, we conduct a series of ablation experiments on the three benchmark datasets. Specifically, consider the following aspects.

1) *w/o SAM-Based Registration (w/o SOR.):* The variant removes the entire SAM-guided registration module, including both affine alignment and label expansion. In this case, the model directly performs change detection based on the original spatial coordinates, without any geometric alignment or label propagation.
2) *w/o Region-Based Propagation via Mask Intersection (w/o RLPMI):* The version discards the label propagation mechanism described in Section III-C, using only sparse ground-truth annotations for training.
3) *w/o SST-MCDNet (w/o SST):* We replace the proposed SST-MCDNet with a plain Mamba-based backbone that processes the spectral dimension independently without joint spatial–spectral modeling.
4) *Sensitivity to Pseudo-Color Generation:* We compare the proposed wavelength-aligned RGB selection with PCA-based dimensionality reduction.

1) Importance of SAM-Based Registration: We first investigate the effect of discarding the registration module, which includes the affine alignment and label propagation components. Compared with the full model, removing this module leads to a noticeable decline in performance across all datasets, as shown in Fig. 8. For instance, on the BAR dataset, OA decreases from 0.9910 to 0.9798, Kappa drops from 0.9812 to 0.9575, and the $F1$-score reduces from 0.9886 to 0.9741. Similar trends are observed for the BAY and Hermiston datasets, where Kappa drops by approximately 0.0145 and 0.09, respectively. These results highlight the importance of the registration module in providing spatially aligned guidance for the change detection network.

2) Impact of Label Expansion: Next, we evaluate the effect of eliminating the label expansion module while retaining the registration process. As shown in Fig. 9, without label expansion, the model loses the ability to propagate sparse annotations to overlapping regions, which decreases the dense supervision available during training. This is particularly reflected in the Kappa metric: for example, on the

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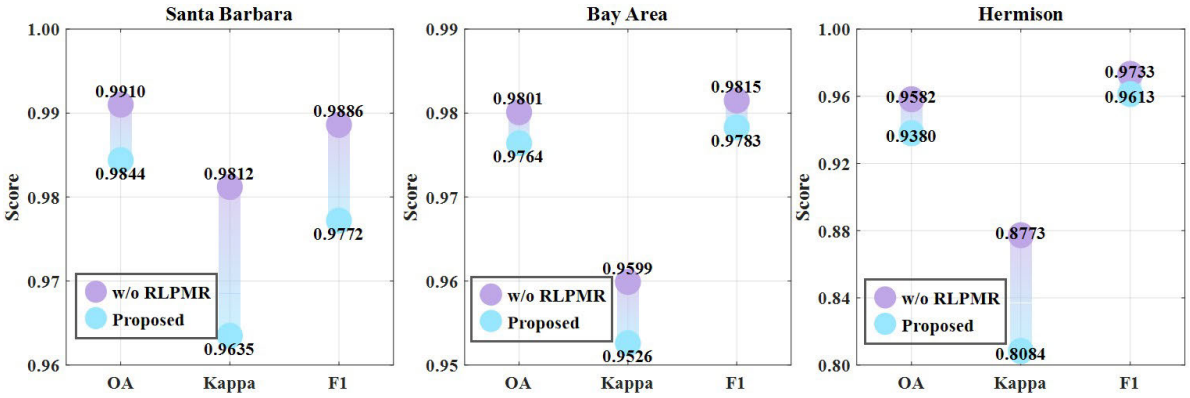


Fig. 9. Performance comparison of the proposed method and its variant, which removes RLPMI on three datasets.

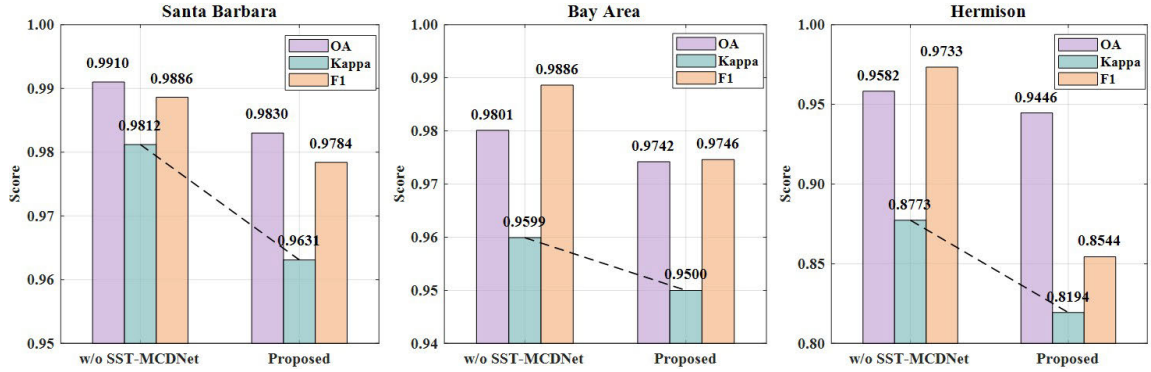


Fig. 10. Performance comparison of the proposed method and its variant, which removes SST-MCDNet on three datasets.

BAR dataset, Kappa drops from 0.9812 to 0.9635, and on BAY it decreases from 0.9599 to 0.9526, while OA and $F1$ show relatively minor reductions. In addition, compared with the results shown in Fig. 8 where both registration and label expansion are removed, and the performance obtained by retaining the registration module is noticeably improved. These results indicate that the registration module effectively mitigates misalignment and ensures structurally consistent feature correspondence. However, the absence of label expansion constrains the model's ability to fully exploit the registered structures for robust change discrimination.

3) Role of SST-MCDNet: We replace the proposed SST-MCDNet with a standard Mamba-based network, while keeping the dual-branch encoder–decoder framework intact. It aims to evaluate the contribution of our spatio-spectral-temporal modeling. As shown in Fig. 10, the results indicate a clear decline in change detection accuracy with the standard Mamba-based network compared with the proposed SST-MCDNet. Specifically, on the BAR dataset, OA decreases from 0.9910 to 0.9830, Kappa from 0.9812 to 0.9631, and $F1$ from 0.9886 to 0.9784. Comparable drops are observed for the BAY and Hermiston datasets. These results indicate that the proposed SST-MCDNet is essential for capturing both local spatial–spectral cues and intertemporal differences.

4) Sensitivity to Pseudo-Color Generation: To verify the robustness of our band-selection strategy, we conducted a sensitivity analysis comparing it with PCA-based dimensionality reduction. As shown in Table IV, the SAM segmentation masks generated by both strategies exhibit high consistency, with an average intersection-over-union (IoU) of 0.8824 across the three datasets. This indicates that the segmentation capability of SAM is primarily driven by spatial boundaries and structural priors rather than specific spectral projections. Consequently, our wavelength-aligned strategy achieves comparable performance to PCA while maintaining lower computational complexity and better interpretability.

TABLE IV
IoU Between Segmentation Results Generated by PCA-Based and RGB Band-Selection Strategies

Dataset	Hermiston	BAY	BAR
IoU	0.8837	0.8745	0.8892

5) Effectiveness of PL Expansion: To verify the effectiveness of the expanded PLs, we analyzed both quantity and accuracy. The initial labels increased substantially—from 1325 to 50 428 for Barbara, 735 to 40 231 for Bay Area, and 740 to 30,650 for Hermiston (USA) (see Table V), providing sufficient training samples for the SST-Mamba network. In terms of accuracy, Barbara and Bay Area datasets were near 100%, with only 24 and 51 errors, respectively, while Hermiston remained above 92%. The PLs also showed low false positive rates and high recall, rarely misclassifying "no-change" pixels and effectively capturing real changes. Overall, the module balanced quantity and quality, generating a large number of reliable PLs with minimal noise.

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TABLE V
CONFUSION MATRIX FOR CROSS-STATISTICS BETWEEN PLS GENERATED BY LABEL EXPANSION MODULE AND GROUND-TRUTH LABELS

Dataset	Barbara		BayArea		Hermiston	
Type / Label Propagation	Changed	Unchanged	Changed	Unchanged	Changed	Unchanged
Total Number	1325 → 50428		735 → 40231		740 → 30650	
Correct	25946	24458	21040	19140	9018	20307
Error	24	0	19	32	781	544

TABLE VI
COMPUTATIONAL EFFICIENCY OF SDCO-MDDN AND OTHER COMPETING METHODS

Method	CVA	RECNN	DADSSFF	DIEFEN	MSDFFN	MFCEN	Proposed
Params (M)	/	3.97	6.14	8.29	26.20	77.11	2.35
FLOPs (M)	/	36.45	313.34	29.22	1253.68	583.60	160.54
Inference time (s)	5.42	9.04	22.12	23.42	56.43	29.45	15.87

6) Computational Complexity and Efficiency Analysis: The evaluation metrics include the change detection network's parameters, FLOPs, and total inference time, covering both the registration module and the detection network. Unlike neural networks, CVA has no trainable parameters or convolutional/fully connected operations, so its complexity cannot be quantified by parameters or FLOPs. As shown in Table VI, SST-MCDNet achieves the highest detection accuracy with only 2.35M parameters, the lowest among all compared methods. Its FLOPs are 160.54M, ranking third, slightly higher than DIFEFEN and RECNN and much lower than the others. With the SAM-guided registration module included, the total inference time is 15.87 s, outperforming most competing methods and only slightly slower than CVA and RECNN.

These ablation studies confirm that each module-SAM-guided registration, label expansion, and SST-MCDNet is indispensable for achieving robust and accurate HSCD, particularly under spatially misaligned conditions.

V. CONCLUSION

In this article, we proposed an HSCD framework that addresses the challenge of spatial misalignment through object-level structural priors. By leveraging segmentation masks generated by SAM, our method connects image registration and sparse label propagation into a unified pipeline, enabling region-consistent supervision even under geometric distortions. We further designed a spatio-spectral-temporal change detection network based on convolution and Mamba modules, which jointly capture spatial, spectral, and temporal dependencies. Extensive experiments on three hyperspectral datasets demonstrate that the proposed method consistently outperforms state-of-the-art baselines in both perfectly aligned and misaligned scenarios. The results highlight the effectiveness of introducing object-level priors and the robustness of our framework in realistic RS applications. In the future, we will extend the framework to tackle HSCD under nonrigid deformations (e.g., vegetation growth, building structural variations, and terrain surface undulation), which are widespread in complex practical scenarios and serve as a natural extension of the focus of the current work on rigid misalignments.

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